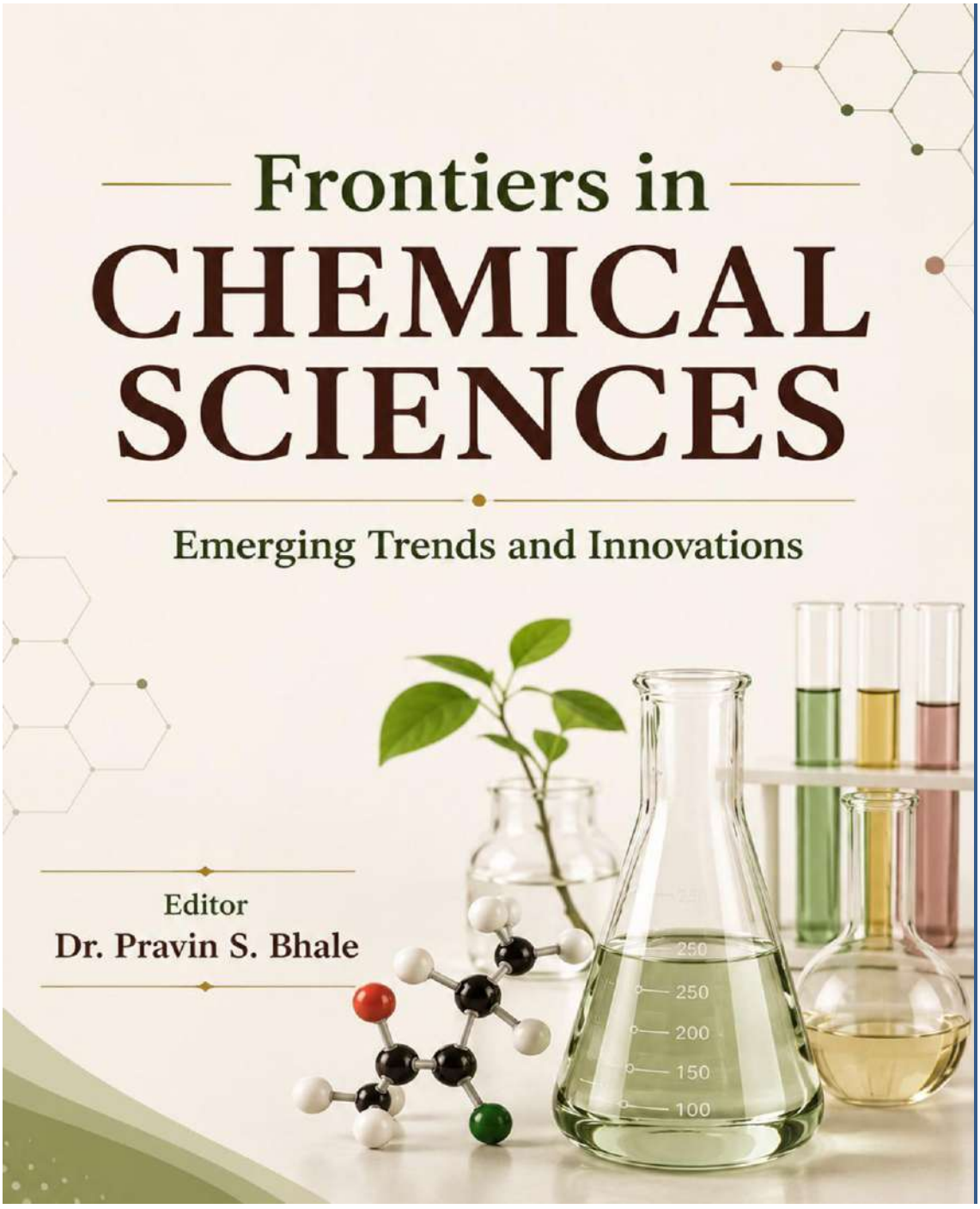
📚 **Book Chapter Publication Achievement**

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EMERGING SMART NANOMATERIALS: ADVANCES IN NANOMEDICINE, SUSTAINABLE TECHNOLOGIES, AND INDUSTRIAL INNOVATIONS

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Abstract

Smart nanomaterials represent a revolutionary class of functional materials engineered at the nanoscale to respond dynamically to external stimuli. This chapter comprehensively reviews the synthesis, characterization, and multifaceted applications of smart nanomaterials across nanomedicine, sustainable technologies, and industrial sectors. Smart nanomaterials exhibit size-dependent properties and stimuli-responsive behaviors that enable unprecedented control over drug delivery, environmental remediation, energy storage, and precision healthcare. We examine synthesis strategies ranging from top-down and bottom-up approaches to green synthesis techniques, alongside advanced characterization methodologies. The clinical translation of nanotheranostics, development of biodegradable nanomaterials, and integration with artificial intelligence exemplify emerging frontiers. Future perspectives emphasize personalized nanomedicine, autonomous nanodevices, and circular economy approaches for sustainable nanotechnology deployment. These developments highlight the growing potential of smart nanomaterials to bridge scientific research with real-world applications, while promoting safer, efficient, cost-effective, and environmentally responsible solutions. Continued interdisciplinary collaboration among researchers, healthcare professionals, engineers, industries, and policymakers will be essential to accelerate innovation, ensure responsible development, and support the successful integration of smart nanomaterials into future healthcare, environmental, energy, and manufacturing systems worldwide.

Keywords: *Smart Nanomaterials; Nanomedicine; Sustainable Technologies; Nanotheranostics; Green Synthesis; Artificial Intelligence; Industrial Innovations.*

1. Introduction to Smart Nanomaterials

1.1 Evolution of Nanotechnology and Nanomaterials

The emergence of nanotechnology as a distinct scientific and technological discipline has fundamentally transformed materials science over the past two decades. Nanotechnology focuses on engineering structures and systems at the nanoscale (1-100 nanometers), where quantum mechanical effects become predominant and novel material properties emerge. The historical trajectory of nanomaterials development began with theoretical predictions by physicists in the 1970s and culminated in practical synthesis techniques and commercial applications by the early 2000s. Carbon nanotubes, fullerenes, and metallic

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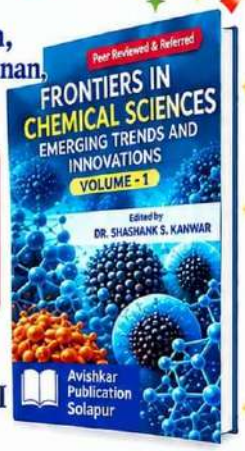
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